Reg. No. | | | | | | | | | | | [18MTSE203]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019
EARTHQUAKE ANALAYSIS AND DESIGN OF STRUCTURES

Time: 03 Hours **Maximum Marks: 100**

Instructions:
- **i. Answer FIVE full Questions. Questions 1& 2 are compulsory**
- **ii. Missing data may be suitably assumed.**
- **iii. Use of IS 1893-2016 is permitted.**

1. a. How earthquakes are classified based on different aspects? **10**

b. Explain tripartite plot of response spectrum and significance of spectral regions. **10**

2. Analyse the building frame for the following data using response spectrum method. **20**

Zone= III, Importance factor=1.0, Structure= OMRF, Soil= Medium, Seismic masses are $M1=M2=M3=368.91X10^3$ kg

$K1=K2=K3=K4= 0.6075X10^9$ N/m.

Ø=	1.00	1.00	1.00	1.00
	1.87	0.91	-0.48	-1.60
	2.48	-0.17	-0.77	1.55
	2.77	-1.07	0.85	-0.85

3. a. Compare between static and dynamic analysis. **06**

b. Discuss the concept of plan and vertical irregularities. **14**

OR

4. a. Explain the concept of containment reinforcement. **10**

b. Discuss the concept of earthquake resistant masonry. **10**

5. a. Discuss the design loads and load combinations in the design of RC buildings for earthquake. **10**

b. Explain the design and ductile detailing of shear walls. **10**

OR

6. a. Write and explain the objectives of earthquake resistant design of structures. **10**

b. Explain the important response characteristics of structures. **10**

7. a. Discuss the performance based seismic engineering methodology. **10**

b. Discuss the concepts of seismic response control. **10**

OR

8. a. Write an overview of linear and non-linear seismic analysis. **10**

b. Compare the response of brittle and ductile systems. **10**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112